# Gratheon.com

## Monitor bees with AI



**Artjom Kurapov**

Founder

# Precise pollination

- Pollination boosts crop yields (+10-30%)
- Farmers lack sufficient amount of pollinators
- Beekeepers providing services to farmers earn 9x more money compared to their honey income
- Demand of pollination grows 2x faster than growth of honeybee colonies



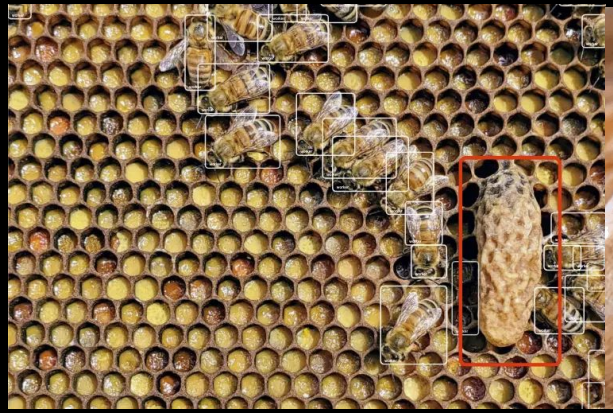
# Human worker efficiency

- Hard physical labour
- Not scalable with amount of beehives
- Driving to remote locations takes time



# Bee colony health observability

- Need regular (weekly) inspections
- Treat against varroa mite infestations
- Predict swarming, track queen presence
- Track colony growth/efficiency, prevent starvation





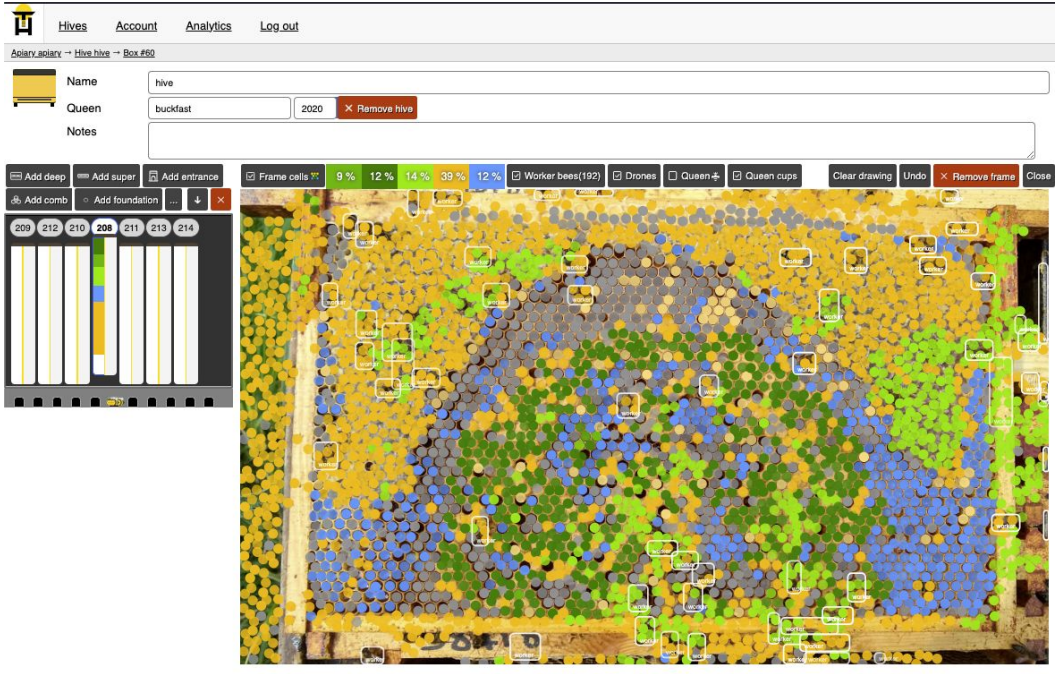
# Vision

app

Software as a service for beekeepers to organize and monitor their apiary

- ☑ AI detections of a frame photo
- ☑ manage apiaries, hives, frames, photos
- ☑ Queen and queen cups
- ☑ Varroa mites
- ☑ Frame cells
- ☑ AI advisor
- ☑ Mobile app
- ☑ Manual inspections

- ▢ Re-train on user data
- ▢ Analytics (graphs) to find correlations/anomalies
- ▢ Alerts



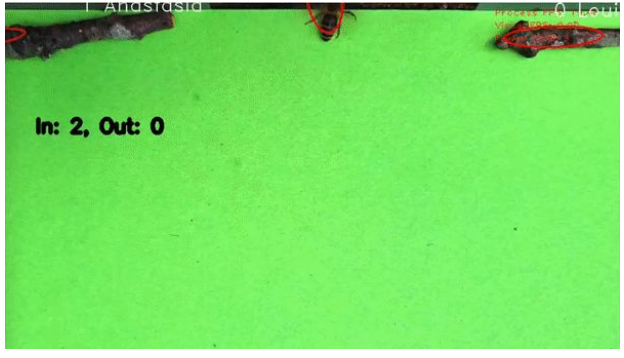
# Vision

app → Entrance Observer

Hive entrance video monitoring / IoT device

- ☑ Video streaming & playback
- ☑ Incoming/Outgoing bee counter

- ▢ Varroa mite detection
- ▢ Foraging analysis
- ▢ Stealing state
- ▢ Hornet detection



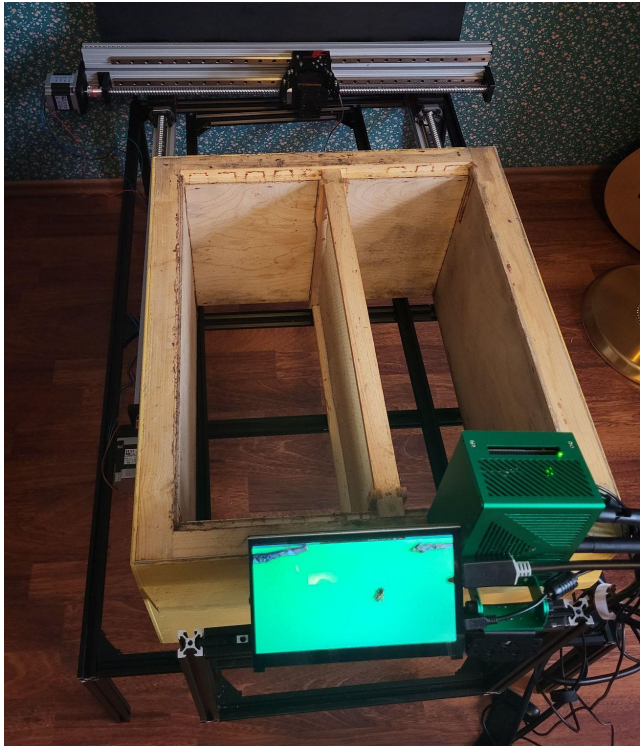



# Vision

app → Entrance Observer → Robot

- Frame extraction mechanism
- 2 cameras
- Robotic beehive. Initially single colony
- Later, moving on rails for inspect multiple hives
- Permanent frame extraction
- Frame movement across hives
- Ventilation control

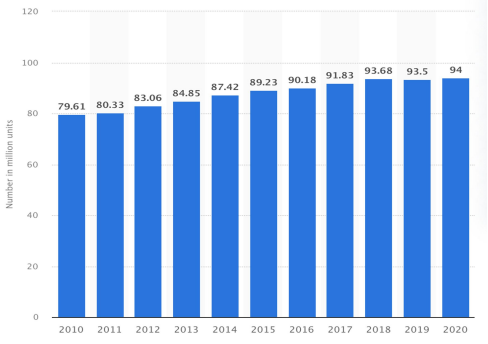
- Move robot on wheels to serve multiple beehives

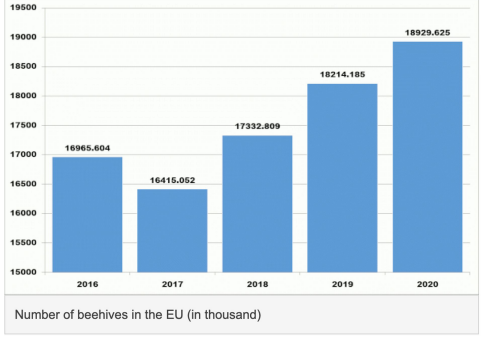


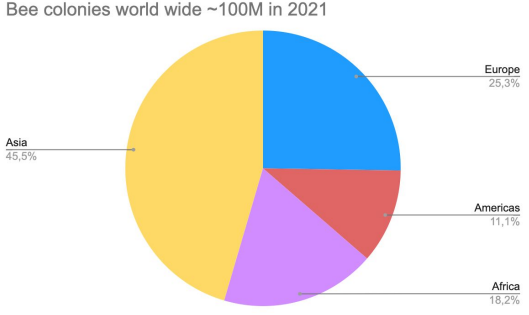
early prototype

# Market

- Target customers - **industrial beekeepers** (B2B, robot)
- Secondary - backyard beekeepers (B2C, web-app, manual inspections)
- Addressable market
  - Estonia - 6425 **apiaries** (PRIA sources). ~ 60% beekeepers have > 25 bee colonies
  - Europe - 620k **beekeepers, 19-25M** colonies
  - World wide - 94M **colonies** in 2020 → 101.6 M **colonies** in 2021



# ⛰️ Market estimation

10% market share * 1M industrial beekeepers * 2 robots * 150 EUR/mo =

5% market share * 2M beekeepers * 15 EUR/mo for web-app = 18M EUR YoY

# Invest

- Raising 35k angel investments
  - Hardware development (Entrance observer)
  - Field testing with local beekeepers
  - GPU hosting cluster + cloud video storage

- Raising 1M pre-seed round for runway: 24 months
  - **Team of 4** + external contractors & beekeepers
  - AI models improvements
  - Robot R&D

**pilot@gratheon.com**

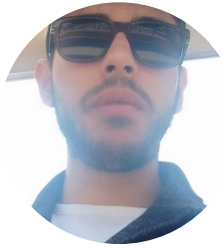
# 🐝 Team



**Artjom Kurapov**
Founding engineer / beekeeper
(ex-Pipedrive, Clarifai)



**Šimon Bilík**
PHD Researcher
System engineer / Beekeeper



**Ahmed Daoudi**
Fullstack engineer volunteer

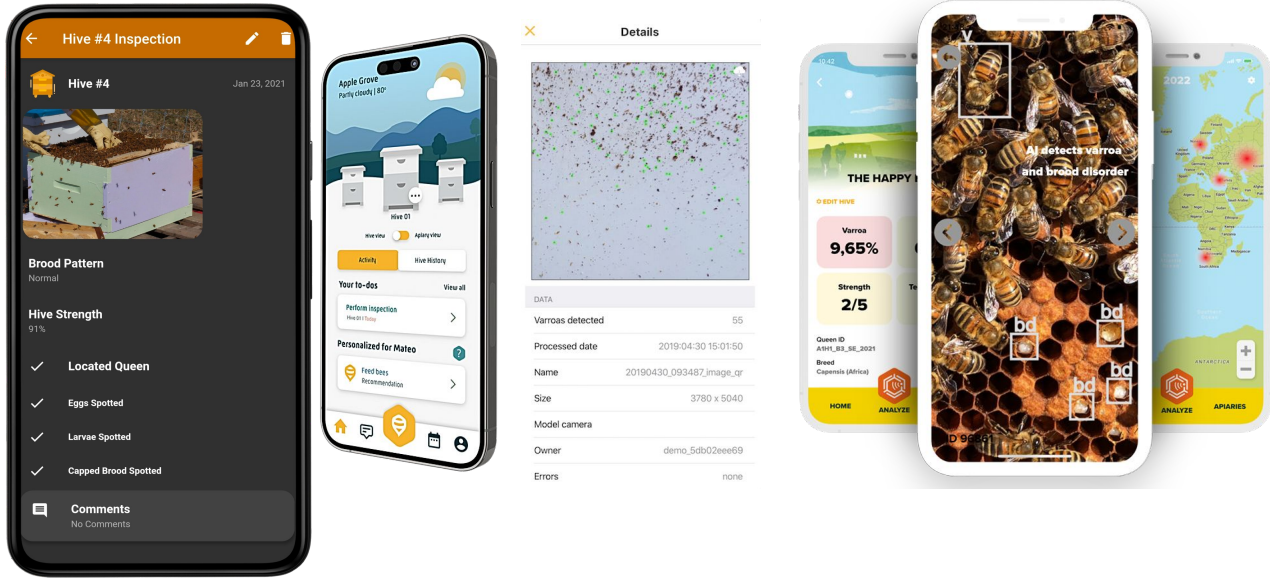
# Pricing model

| Community | Essential | Professional |
| --- | --- | --- |
| free | **15** EUR / month<br>2 weeks trial, annual billing | **5** EUR per beehive per month<br>+ **10** EUR per user per month |
| 5 hives max | ✅ More AI detection features<br>✅ Timeline<br>✅ Sharing<br>✅ Alerting<br>✅ Telemetry API | (All of Essential plan, plus)<br>✅ Analytics + Demographic breakdown<br>✅ Multiple users<br>✅ Inventory management<br>✅ Video streaming playback |



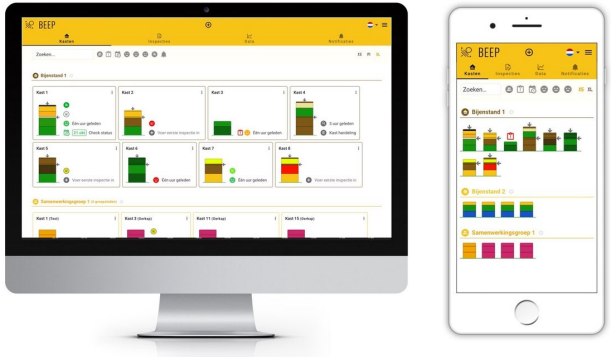
# Competition - Data organizer apps

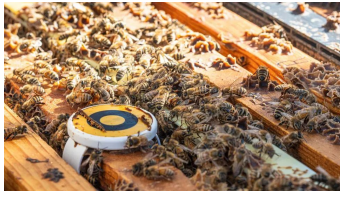
- BeeScanning
- ApiZoom
- HiveTracks
- HiveBloom
- BeeQueenDetector
- apimanager
- apiary book



# IoT - analog data (audio, humidity, temperature)

- beep.nl - opensource
- broodminder.com
- beelab.se
- intelligenthives.eu
- beehivemonitoring.com
- solutionbee.com
- beehivemonitoringusa.com
- osbeehives.com
- beesage.co





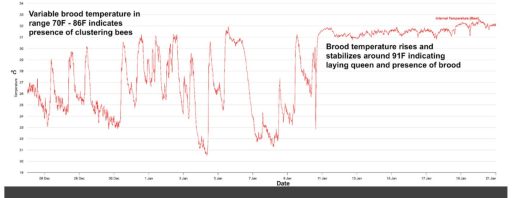



Fig. 2: Using Brood temperature to detect onset of laying queen in late winter/early spring

# Vision AI and hardware

- beemate.buzz - counts bees (Australia, ~ 340 eur)
  - closed system
- beewise.ag - robotic multi-colony container hive. Moves and analyzes individual frames
  - big, many colonies



